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PRACTICAL - 12

Object Oriented Programming (CEUE203)

AIM:

To implement and use different collection classes available in Java such as ArrayList, LinkedList, HashSet, HashMap and TreeMap and demonstrate sorting using Comparator/Comparable, and concurrent collections.

PART A - PRACTICE PROGRAMS (HOUR 1)

1. Word Frequency

Split a paragraph into words, count each in a HashMap, and print words sorted by frequency (highest first) using a Comparator.

Program :

```
import java.util.*;

public class WordFrequency {
    public static void main(String[] args) {
        String text = "Java is easy Java is fun Java is powerful";
        String[] arr = text.split" ";
        HashMap<String,Integer> map = new HashMap<>();

        for(String word : arr)
            map.put(word, map.getOrDefault(word, 0)+1);

        ArrayList<Map.Entry<String,Integer^`list = new ArrayList<>
        (map.entrySet());
        list.sort((a,b) -> b.getValue()-a.getValue());

        System.out.println("Word Frequencies : ");

        for(Map.Entry<String,Integer> e : list)

            System.out.println(e.getKey()+ " : " +
            e.getValue());

    }
}
```

Output :

```
Word Frequencies :
Java : 3
is : 3
easy : 1
fun : 1
powerful : 1
```

2. Unique Visitors

From a list of login names (with repeats), use Set classes to count unique users and keep first-seen order. Print unique count and first three names.

Program :

```
import java.util.*;
public class UniqueVisitors {
    public static void main(String[] args) {
          String[] logins = {“jay”,“neel”,“jay”,“om”,“neel”,“khan”};

          HashSet<String> set = new HashSet<>();
          LinkedHashSet<String> ordered = new LinkedHashSet<>();

          for(String name  : logins){
                set.add(name);
                ordered.add(name);
          }
          System.out.println(“Unique Count : " + set.size());
          System.out.println(“First Three : ");
          int c = 0;
          for(String n : ordered){
                if(c==3) break;
                System.out.println(n);
                c++;
          }
    }
}
```

Output :

```
Unique Count : 4
First Three :
jay
neel
om
```

3. Leaderboard (Advanced – Home)

Build a leaderboard from player scores kept in a Map, and print top-N sorted by score (ties broken by name).

Program :

```
import java.util.*;
public class Leaderbourd {
    public static void main(String[] args) {
          HashMap<String,Integer> scores = new HashMap<>();
          scores.put(“Jay",120);
          scores.put(“Khan",120);
          scores.put(“Neel",250);
          scores.put("Om",180);
          scores.put(“Zyan",300);
          ArrayList<Map.Entry<String,Integer>> list = new ArrayList<>(scores.entrySet());
          list.sort((a,b)->{
             int cmp = b.getValue()-a.getValue();
             if(cmp==0) return a.getKey().compareTo(b.getKey());
             return cmp;

          });

          System.out.println(“Top Leaderboard : ");
          list.forEach(e -> System.out.println(e.getKey()+" :"+ e.getValue()));
    }
}
```

Output :

```
Top Leaderboard :
Zyan : 300
Neel : 250
Om : 180
Jay : 120
Khan : 120
```